

Computer Networks

Lecture 32 Application Layer

Content



- WWW



The World Wide Web (WWW) is a large collection of information resources that are linked together and can be accessed from different parts of the world through the Internet.

It provides a simple and user-friendly way to access information in the form of web pages, which may contain text, images, audio, video, and hyperlinks.

The WWW offers:

- **Flexibility:** Information can be presented in different formats.
- **Portability:** Web resources can be accessed from different devices and platforms.
- **User-friendly access:** Users can easily navigate through web pages using browsers and hyperlinks.

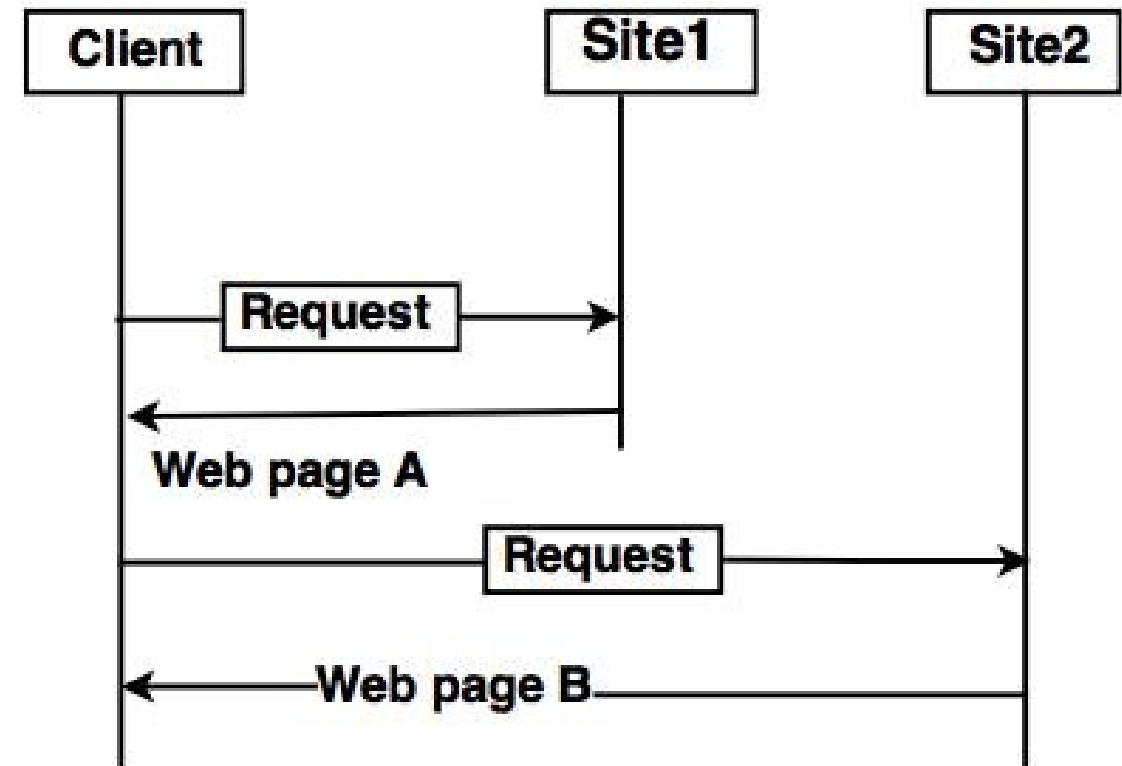
WWW Architecture

The WWW today is a distributed client server service, in which a client using a browser can access a service using a server.

The service provided is distributed over many locations called sites

Each website contains one or more **web pages**

When a user clicks a link on that page, the browser sends a new request to the linked site, and the new page is retrieved.



Architecture of WWW

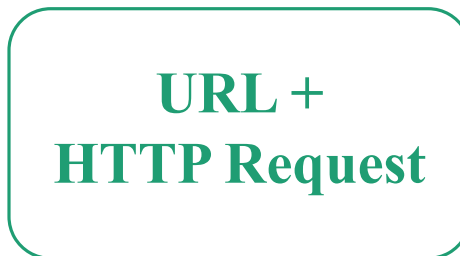
Retrieve Web Page

A browser requests a page, and the web server sends it back.

1. User enters a URL
or clicks a link



2. Browser sends an
HTTP/HTTPS request



3. Server returns the
requested web page



The received page may contain hyperlinks to other pages. When a link is clicked, the browser sends a new request to the corresponding site.

Components of WWW

Client

The user side of the web. A browser such as Chrome, Firefox or Edge requests web pages and displays them to the user.

Server

A computer that stores websites and web pages. It receives client requests, locates the requested file, and sends it back.

URL

Uniform Resource Locator. It is the address that tells the browser where a particular web page or resource is located.

Cookies in WWW

Cookies are supporting elements used during web browsing.

What are cookies?

Cookies are small pieces of data stored by the browser on the user's device. They help websites remember information about the user.

Used for

- Login/session details
- User preferences
- Shopping cart information
- Personalized browsing experience

Cookies are not a core component like client, server or URL. They support the WWW by maintaining state and improving user experience.

Web Document

The documents in the WWW can be grouped into three broad categories:

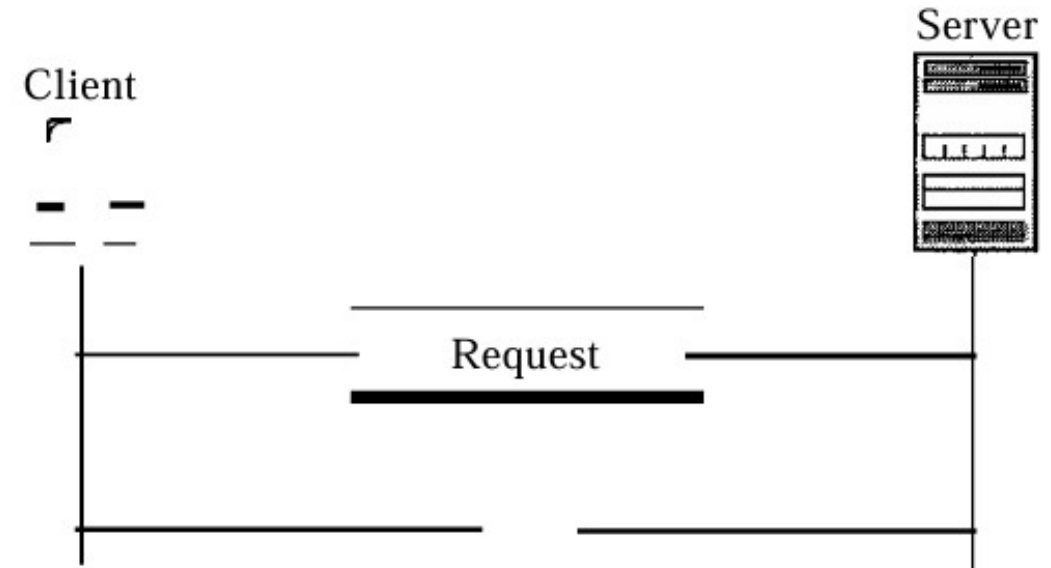
- static
- dynamic
- and active

The category is based on the time at which the contents of the document are determined.

Web Document: Static Document

Static documents are fixed-content documents that are created and stored in a server. The client can get only a copy of the document.

The contents in the server can be changed, but the user cannot change them. When a client accesses the document, a copy of the document is sent.



Static HTML document

Web Document: Dynamic Document

A dynamic document is a web document whose content can change automatically depending on the user, request, time, or data available.

Example: When you open a shopping website:

- The homepage may show products based on your search history.
- Your cart shows only the items you selected.
- Your account page displays your personal details.

In the World Wide Web, dynamic documents are usually created using:

- server-side programs
- databases
- scripts
- user input

Web Document: Active Document

An active document is a web document that contains a program or script that can run on the client side, usually inside the user's browser.

Example: When you open a web page and it shows:

- live form validation
- automatic updates without reloading the full page
- interactive maps

then that page uses an active document.

An active document is a web page that contains executable code, which runs in the browser and makes the page interactive.